

Assignment: Implement Stack and Queue

1. Implement Stack Using Array

Requirements:

- Implement a stack using both arrays.
- Include the following operations:
 - push: Add an element to the stack.
 - pop: Remove the top element from the stack.
 - peek: Retrieve the top element without removing it.
 - isEmpty: Check if the stack is empty.
 - size: Return the number of elements in the stack.

2. Implement Stack Using Linked List

Requirements:

- Implement a stack using linked list.
- Include the following operations:
 - push: Add an element to the stack.
 - pop: Remove the top element from the stack.
 - peek: Retrieve the top element without removing it.
 - isEmpty: Check if the stack is empty.
 - size: Return the number of elements in the stack.

3. Implement Queue Using Array

Requirements:

- Implement a queue using Array
- Include the following operations:
 - enqueue: Add an element to the queue.
 - dequeue: Remove the front element from the queue.
 - peek: Retrieve the front element without removing it.
 - isEmpty: Check if the queue is empty.

- size: Return the number of elements in the queue.

4. Implement Queue Using Linked List

Requirements:

- Implement a queue using linked list.
- Include the following operations:
 - enqueue: Add an element to the queue.
 - dequeue: Remove the front element from the queue.
 - peek: Retrieve the front element without removing it.
 - isEmpty: Check if the queue is empty.
 - size: Return the number of elements in the queue.

5. Implement Stack Using One Queue

Requirements:

- Implement a stack using one Queues.
- Include the following operations:
 - push: Add an element to the stack.
 - pop: Remove the top element from the stack.
 - peek: Retrieve the top element without removing it.
 - isEmpty: Check if the stack is empty.
 - size: Return the number of elements in the stack.

6. Implement Stack Using Two Queue

Requirements:

- Implement a stack using two queues.
- Include the following operations:
 - push: Add an element to the stack.
 - pop: Remove the top element from the stack.
 - peek: Retrieve the top element without removing it.
 - isEmpty: Check if the stack is empty.
 - size: Return the number of elements in the stack.

7. Implement Queue Using One Stack

Requirements:

- Implement a queue using one stack.
- Include the following operations:
 - enqueue: Add an element to the queue.
 - dequeue: Remove the front element from the queue.
 - peek: Retrieve the front element without removing it.
 - isEmpty: Check if the queue is empty.
 - size: Return the number of elements in the queue.

8. Implement Queue Using Two Stack

Requirements:

- Implement a queue using two stacks.
- Include the following operations:
 - enqueue: Add an element to the queue.
 - dequeue: Remove the front element from the queue.
 - peek: Retrieve the front element without removing it.
 - isEmpty: Check if the queue is empty.
 - size: Return the number of elements in the queue.

Submission Guidelines

1. Raise a separate Pull Request (PR) for each assignment.
2. In the PR description, include:
 - a. A brief overview of the implementation.
 - b. Time complexity analysis.
 - c. Space complexity analysis.
3. Ensure the code adheres to clean coding principles and best practices.
4. Test the implementation thoroughly, including edge cases.